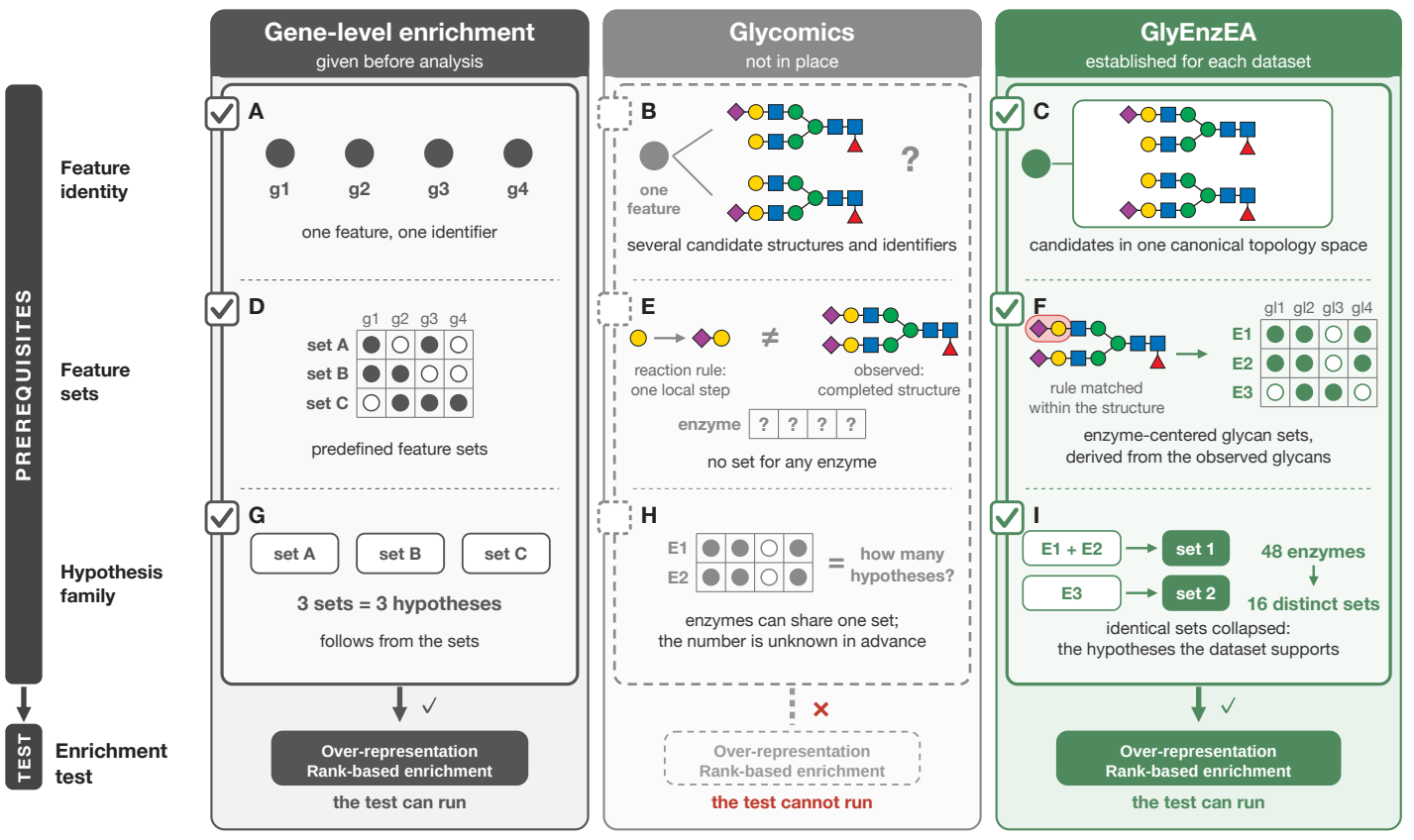

PREREQUISITES
TEST
Feature identity
Feature sets
Hypothesis family
Enrichment test
Gene-level enrichment
given before analysis
A
g1
g2
g3
g4
one feature, one identifier
D
g1 g2 g3 g4
set A
set B
set C
predefined feature sets
G
set A
set B
set C
3 sets = 3 hypotheses
follows from the sets
Over-representation
Rank-based enrichment
the test can run
Glycomics
not in place
B
one feature
?
several candidate structures and identifiers
E
reaction rule:
one local step
≠
observed:
completed structure
enzyme ? ? ? ?
no set for any enzyme
H
E1
E2
= how many hypotheses?
enzymes can share one set;
the number is unknown in advance
Over-representation
Rank-based enrichment
the test cannot run
GlyEnzEA
established for each dataset
C
candidates in one canonical topology space
F
rule matched
within the structure
gl1 gl2 gl3 gl4
E1
E2
E3
enzyme-centered glycan sets,
derived from the observed glycans
I
E1 + E2
set 1
E3
set 2
48 enzymes
16 distinct sets
identical sets collapsed:
the hypotheses the dataset supports
Over-representation
Rank-based enrichment
the test can run